Advanced Trophic Cascades and Metabolic Energetics in Ecosystem Simulation: A Thermodynamic Monad Approach

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Abstract

Ecosystem simulation models often struggle with artificial energy generation and mass non-conservation due to decoupled metabolic accounting. In this paper, we detail the implementation of Sprint 029 for the **Web of Life** simulation engine (`https://github.com/pascalranoroarijaona/WebOfLife`), which introduces strict thermodynamic tracking and multi-tiered trophic cascades. We formalize energy dissipation via Kleiber's Law, stoichiometric mass conservation across trophic boundaries, and dynamic predator-prey functional responses. Furthermore, we present an executable state vector validation framework that enforces the First and Second Laws of Thermodynamics across all biogeochemical monad state transitions.

1 Introduction and Sprint Objectives

Traditional artificial life simulations frequently abstract away fundamental physical laws, permitting energy inflation and unregulated organismal proliferation. Sprint 029 bridges the gap between theoretical systems ecology and computational execution by embedding rigorous thermodynamic constraints directly into the simulation monads.

The core objectives of Sprint 029 are:

1. Enforcing strict First Law mass-energy conservation across closed pod boundaries.
2. Integrating Second Law entropy generation checks and absolute temperature positivity constraints.
3. Implementing organismal metabolic engines governed by Kleiber's Law ($BMR = B_0 M^{3/4}$).
4. Deploying system-level telemetry via the `TrophicCascadeMonitor`.

2 Thermodynamic Framework

2.1 First Law: Conservation of Mass and Energy

For any monad state transformation $\mathcal{M} : \Gamma_t \to \Gamma_{t+\Delta t}$, the total mass of biochemical stocks (Carbon, Nitrogen, Phosphorus, Water) and internal energy E must satisfy exact balance preservation bounds:

$$\Delta M_{\text{system}} = \sum_i \Delta m_i = 0 \quad \text{(Closed Pod Boundary)} \tag{1}$$

$$\Delta E_{\text{system}} = Q_{\text{net}} - W_{\text{work}} \tag{2}$$

where individual material stocks are bounded by $m_i \geq 0$.

2.2 Second Law: Entropy and Temperature Constraints

Entropy S and absolute temperature T must obey strict physical inequalities during all computational steps:

$$S \geq 0, \quad T > 0 \tag{3}$$

Violations of these conditions indicate unphysical states and trigger immediate execution halts.

3 Stock Transfer and Process Matrix

Table 1 summarizes the biogeochemical stock transformations and their respective thermodynamic deltas.

Table 1: Biogeochemical Monad Stock Transitions and Thermodynamic Signatures

Process / Monad Step	**Input Stock (I)**	**Output Stock (O)**	**Mass Delta (Δm)**	**Entropy Delt**
Photosynthesis	CO_2, H_2O, Q_{solar}	$C_6H_{12}O_6, O_2$	$\sum \Delta m = 0$	$\Delta S_{\text{univ}} \geq 0$
Respiration	$C_6H_{12}O_6, O_2$	CO_2, H_2O	$\sum \Delta m = 0$	$\Delta S_{\text{univ}} \geq 0$
Hydrological Cycle	$H_2O_{(l)}$	$H_2O_{(g)}$	$\Delta m_{H_2O} = 0$	$\Delta S_{\text{univ}} \geq 0$

4 Executable Runtime Validation

To guarantee compliance, state transitions are wrapped using validator functions:

```
public static wrapMonadStep(stepFn: MonadStepFunction): MonadStepFunction {
  return (vector: ThermodynamicStateVector): ThermodynamicStateVector => {
    this.validateStateVector(vector);
    const nextVector = stepFn(vector);
    this.validateStateVector(nextVector);
    return nextVector;
  };
}
```

5 Conclusion

Sprint 029 establishes a physically grounded computational ecology within the Web of Life framework. By enforcing thermodynamic invariants at the monad level, we eliminate artificial energy generation and enable realistic ecosystem dynamics.

Source Code & Repository: `https://github.com/pascalranoroarijaona/WebOfLife`